Major Project Synopsis

On

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GUIDELINES FOR MAJOR PROJECT SYNOPSIS

ABSTRACT

The proposed project, **"Digital Platform for Centralized Alumni Data Management and Engagement,"** aims to develop a centralized web-based system for managing alumni information and establishing continuous interaction between alumni, students, and educational institutions. In many institutions, alumni information such as contact details, academic records, employment information, achievements, and professional updates is maintained in scattered spreadsheets, emails, social-media groups, or outdated databases. This makes it difficult to maintain accurate records and effectively communicate with alumni.

The proposed system provides a unified platform where alumni can register, maintain their professional profiles, search for other alumni, participate in institutional events, and share career opportunities. Students can use the platform to discover alumni, explore career opportunities, and interact with experienced professionals. Administrators can verify alumni registrations, manage events and job postings, monitor platform activities, and analyze alumni engagement through dashboards and reports.

The system will be developed using a modern web architecture consisting of **React with Vite** for the frontend, **Spring Boot** for backend services, and **MongoDB** for data management. RESTful APIs will facilitate communication between the frontend and backend. Authentication, role-based access control, directory search, event management, job posting, notifications, and audit logging will be incorporated to ensure functionality and security.

The proposed platform will improve the accuracy, accessibility, and maintainability of alumni records while creating a structured channel for long-term alumni engagement and institutional networking.

1. INTRODUCTION

Alumni are an important part of an educational institution's long-term ecosystem. They can contribute through mentorship, career guidance, professional networking, recruitment opportunities, academic collaboration, and institutional development. However, maintaining an effective relationship with alumni becomes difficult when alumni information is distributed across multiple spreadsheets, emails, messaging applications, and manually maintained records.

Traditional alumni management approaches generally focus on maintaining basic contact information and sending occasional communications. Such systems often lack integrated features for profile management, alumni discovery, event participation, job opportunities, notifications, and engagement analytics. As the number of alumni increases, manually maintaining such information becomes inefficient and difficult to scale.

The proposed project addresses these challenges by developing a **centralized digital alumni management and engagement platform**. The system will provide separate functionalities for students, alumni, and administrators. Alumni will be able to maintain their professional information and connect with the institutional community. Students will be able to search the alumni directory and access career-related opportunities. Administrators will be responsible for verification, content management, event management, and monitoring.

The project follows a modern web application architecture using React, Spring Boot, MongoDB, and REST APIs. The platform is intended to transform alumni management from a fragmented record-keeping process into an organized digital ecosystem that supports continuous interaction and collaboration.

2. PROBLEM DOMAIN

Educational institutions often lack a reliable and centralized mechanism for maintaining updated alumni information. Alumni records may be distributed among spreadsheets, documents, emails, and informal messaging groups. Consequently, institutions may lose track of alumni career progress, contact information, achievements, and professional expertise.

The major problems addressed by the proposed system are:

- Scattered and outdated alumni information.
- Difficulty in searching and accessing alumni records.
- Lack of structured communication between alumni, students, and institutions.
- Limited mechanisms for alumni event management and participation.
- Lack of a centralized platform for sharing job opportunities.
- Difficulty in verifying and maintaining authentic alumni profiles.
- Limited analytics for measuring alumni engagement.

Objectives:

1. To centralize alumni information in a structured database.
2. To provide secure role-based access to students, alumni, and administrators.
3. To enable efficient alumni directory search.
4. To facilitate events, job opportunities, and notifications.
5. To improve long-term alumni–student–institution engagement.
6. To provide administrators with analytics and activity monitoring.

3. SOLUTION DOMAIN

The proposed solution is a **centralized web-based Alumni Management and Engagement Platform** designed to provide a single digital environment for alumni, students, and administrators.

The system provides **CSV/Excel bulk import functionality**, allowing administrators to upload existing institutional alumni data. Imported records are stored as centralized alumni records in the MySQL database. These records can exist independently of user accounts, allowing an alumni record to remain available before the alumnus claims it. When an alumnus joins the platform, the system provides a claim workflow through which the individual searches for their existing record and initiates the claiming process. The server performs re-verification before associating the record with the user's account, thereby reducing the possibility of duplicate or incorrect associations.

The backend follows a structured **Spring Boot REST API architecture**, while React with Vite provides the user interface. The MySQL database maintains the relational structure of the application, including the relationships between users, alumni records, events, jobs, mentorship requests, notifications, and other entities.

4. SYSTEM DOMAIN

The proposed system will be implemented as a modern full-stack web application. **React with Vite** will be used for frontend development because it provides a component-based architecture, fast development environment, and responsive user interfaces. **Spring Boot** will be used for backend development because it provides a structured framework for developing scalable RESTful APIs and implementing business logic.

MySQL is selected as the primary database because the application contains strongly related entities and requires relational relationships, foreign keys, constraints, and controlled status values. The database consists of the project's backend entities and maintains data integrity through relational design.

The development environment will include **Visual Studio Code**, **Java Development Kit (JDK)**, Node.js, npm, Git/GitHub, and MySql Modern web browsers will be used for testing. The system can be deployed on a cloud platform to make the application accessible through the internet.

The minimum hardware requirement will be a standard computer with adequate RAM, storage, and internet connectivity. These technologies are selected because they are widely used, scalable, maintainable, and suitable for developing a modern web-based information management system.

5. APPLICATION DOMAIN

The proposed platform can be used by **universities, colleges, departments, and educational institutions** to maintain centralized alumni records and facilitate continuous interaction with their alumni community.

The platform can support alumni directories, professional networking, mentorship, institutional events, job and internship opportunities, announcements, and alumni engagement analytics. Students can benefit from career guidance and professional connections, while alumni can remain connected with their institution and peers.

The system can later be extended with features such as **mobile applications, email/SMS integration, AI-based alumni recommendations, mentorship matching, donation management, and advanced analytics**.

6. EXPECTED OUTCOME

The expected outcomes of the proposed project are:

1. Centralized and structured management of alumni information.
2. Secure role-based access for students, alumni, and administrators.
3. Faster searching and filtering of alumni records.
4. Improved alumni–student and alumni–institution communication.
5. Centralized management of events and job opportunities.
6. Administrative dashboards for monitoring alumni engagement.
7. Improved accuracy, accessibility, and maintainability of alumni data.
8. A scalable foundation for future alumni-related services.

7. REFERENCES

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